Lesson plan · for the teacher

Keep it running and secure

IoT · Level 4 · Lesson 3

Time
40 min

Materials
The board and chalk, or large paper and markers; Sticky notes or slips of paper; A one-page guide template: what it does, how to check it, the battery, whom to call; Optional: a real sensor node, or a photo of one, to point at its parts

Learning goals

- Plan maintenance: batteries, cleaning, recalibration, and a heartbeat that shows a silent failure.
- Secure devices: new passwords, updates, a separate network, only the data needed.
- Hand the system over so it outlives the project team.

Activity: Failure hunt

1. Warm-up (5 min): Tell the story of the garden sensor that showed 40% every day of the holidays while the seedlings died. Ask: how could we have known?
2. Teach (8 min): The upkeep rota (batteries, cleaning, recalibration), the heartbeat and last-seen time, the Mirai story and the security habits, and the handover.
3. Hunt (12 min): In groups, pupils list ways the project could fail. For each: how we would notice, who fixes it, and how to prevent it.
4. Share (7 min): Each group adds its best three to a class table on the board. Circle the silent failures: the ones nobody would notice without a heartbeat.
5. Guide (8 min): Write the device's one-page guide together: what it does, how to check it's alive, how to change the battery, whom to call.

Teach defence only: never try passwords on devices you don't own. Before class, find out who looks after the school's router, so the guide names a real person.

Discussion

- Which failure would your school notice last, and why?
- Where should the school keep the project's passwords?
- Who in the year below could join the team now?

Exit ticket

- Name one silent failure, and say how a heartbeat would reveal it.

Try it in class tomorrow
Failure hunt: Pupils list everything that could stop your sensor project: a flat battery, mud, theft, a lost password, a holiday, an attacker. For each, they write how you would notice, who fixes it, and how to prevent it. Then the class writes the device's one-page guide together.

Worksheet · for pupils

Keep it running and secure

IoT · Level 4 · Lesson 3

Name _____ Class _____ Date _____

1 **Ms Wanjiru runs the school's garden sensors and moves to another school next month. Everything works, and only she knows the passwords. What should she do first?**

- ◯ Write a guide, give the school the passwords, train two people
- ◯ Share one simple password with everyone, so nobody gets locked out
- ◯ Nothing yet: it works now, so it will keep working

2 **A garden node runs on two AA batteries rated 2,000 mAh and averages 2 mA. Plan with 70% of the label.**

Usable capacity: _____ mAh. That lasts about _____ hours, a little over four weeks, so change the batteries every three weeks.

Word bank: 1,400 · 700 · 2,000 · 1,000

3 **The garden dashboard has shown 38% soil moisture, the same number, since Friday. Which could explain it?**

- ☐ The battery is flat, and the screen shows the last reading
- ☐ It hasn't rained, so the moisture can't change
- ☐ A wire has come loose
- ☐ The SIM's prepaid data has run out
- ☐ The sensor has settled, so readings stay steady

4 **Match each habit to the failure or attack it prevents.**

Changing default passwords	• • • •	A dead sensor going unnoticed
Installing updates	• • • •	A leak exposing data you never needed
A separate network for devices	• • • •	A hacked sensor reaching the school's records
Collecting only the data you need	• • • •	Attackers using a known, already-fixed weakness
A heartbeat	• • • •	Malware like Mirai logging in with factory passwords

5 **Put the steps of the 2016 Mirai attack in order, to see where defences break it.**

- [ ] It used them all at once to flood big websites with traffic
- [ ] It installed itself on every device that let it in
- [ ] It tried about 60 factory passwords on each
- [ ] It scanned the internet for cameras and routers

✎ **Who at your school could keep your project running if you left tomorrow? What would they need from you?**